IE643 Lecture 3

Recap:

Supervised learning with perceptron.

$$D = \{(x^i, y^i)\}_{i=1}^{N<\infty}, \quad x^i = \begin{bmatrix} x_1^i \\ x_2^i \\ \vdots \\ x_d^i \end{bmatrix} \in \mathbb{R}^d := X$$

Binary Categorization/Classification

$$y^i \in \{+1, -1\} =: Y$$

h: parametrized function

$$h : X \to Y$$

Perceptron Neuron



Figure 1. A perceptron neuron — inputs $x_1, x_2, \dots, x_d$ each with weight $w_1, w_2, \dots, w_d$, feeding into a dot-product sum $\sum_{j=1}^{d} w_j x_j$. The output branch splits: $\geq \theta \to +1, < \theta \to -1$.

$$\tilde{W} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_d \\ \theta \end{bmatrix}$$

$h_w \equiv F$ — parametric map learned by perceptron

Decision Rule

$$\tilde{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_d \\ -\theta \end{bmatrix}, \quad \tilde{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \\ 1 \end{bmatrix}$$

$$\sum_{j=1}^{d} w_j x_j \geq 0 \ (\geq 0) \quad \text{decision rule}$$

$$\langle \vec{w}, \vec{x} \rangle - \theta \geq 0$$
$$\langle \vec{w}, \vec{x} \rangle - \theta < 0$$

$$\langle \tilde{w}, \tilde{x} \rangle \geq 0$$
$$\langle \tilde{w}, \tilde{x} \rangle < 0$$

$$\tilde{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix}, \quad \tilde{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

Perceptron Prediction

$$h_{\tilde{w}}(x) = \begin{cases} < \tilde{w}, \tilde{x} > \geq 0 \\ < 0 \end{cases}$$

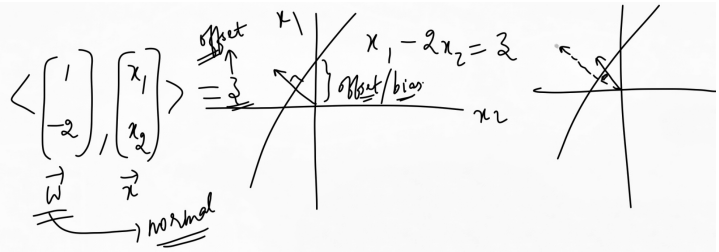


Figure 2. A line $x_1 - 2x_2 = 3$ (offset/bias) plotted on axes x_1, x_2, with a point (x_1, x_2) marked near the line, showing its projection onto the normal vector $\vec{n}$ of the line.

$$\langle \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rangle = 3$$

$\vec{w} \leftarrow$ normal

Train a perceptron:

$$D = \{(x^t, y^t)\} \quad \text{(online learning)}, \qquad t = 1, 2, 3, \ldots$$

$\rightarrow \vec{w}^0 = \vec{0}$

$\rightarrow t = 0, 1, 2, 3, \ldots$

$\rightarrow$ receive a sample $(x^t, y^t) \in D$

$\rightarrow \hat{x}^t = \begin{bmatrix} x^t \\ 1 \end{bmatrix}$

$\rightarrow \hat{y}^t = \text{Perceptron}(\hat{x}^t, \vec{w}^t)$ [Prediction]

Update rule:

if $\hat{y}^t == y^t$: no update, $\vec{w}^{t+1} = \vec{w}^t$ else: $\vec{w}^{t+1} = \vec{w}^t + y^t \hat{x}^t$

$$x_1 + x_2 = 4$$

$$\vec{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \langle \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rangle = 4 = 0$$

$$\text{sign}(z) = +1, z \geq 0$$
$$= -1, \text{otherwise}$$

$$\tilde{w} = \begin{bmatrix} 1 \\ 1 \\ -4 \end{bmatrix}, \quad (x, y) \rightarrow ((2, 1), +1)$$

$$\tilde{x} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{w} = \tilde{w} + y\tilde{x} = \begin{bmatrix} 1 \\ -4 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\text{Perceptron}(\tilde{x}) = \text{sign}(\langle \tilde{w}, \tilde{x} \rangle) = -1$$

Online Learning

$$3x_1 + 2x_2 = 3 \rightarrow \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \rightarrow \text{sign} = +1$$

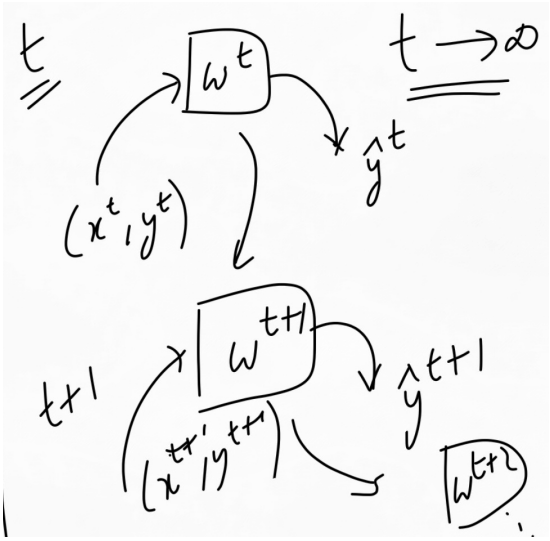


Figure 3. A chain of boxes labeled w^t, (x^t, y^t), w^{t+1}, (x^{t+1}, y^{t+1}), $\hat{y}^t$, $\hat{y}^{t+1}$, w^{t+2}, illustrating the online learning process over time.

Online learning

$t \to \infty, w^t \to w^{t+1}$

finite-sample setup

N samples.

$\to$ predicting "most" no. of samples correctly $\to$ control the fluctuation

Assumption: on the data (linear separability assumption)

?? $\equiv$

Based on this assumption, after a certain t no mistakes are made by perceptron.

Hyperplane:

A set of points $x \in \mathbb{R}^d$ for which there exist a non-zero $w \in \mathbb{R}^d$ and a scalar $b \in \mathbb{R}$ s.t. $\langle w, x \rangle = b$ holds.

Halfspaces

$\langle w, x \rangle - b \geq 0$ $\langle w, x \rangle - b < 0$

two halfspaces associated with $\langle w, x \rangle - b = 0$

Separability Assumption

No updates happen after some finite t

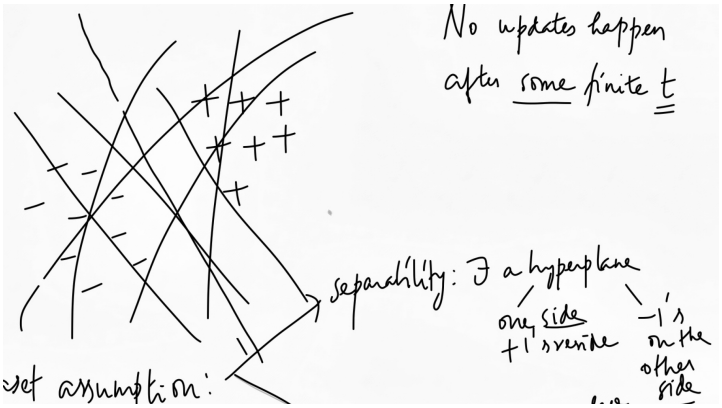


Figure 4. A scatter plot of overlapping "+" and "-" data points with many crossing lines (hyperplanes) drawn through them.

Dataset assumption:

> separability: If a hyperplane one side +'s on one side -'s on the other side

separation with some margin

Linear separability assumption:

$$D = \{(x^t, y^t)\}_{t=1,2,\ldots} \quad (x^t \in \mathbb{R}^d, y^t \in \{+1, -1\})$$

is called linearly separable if there exists a non-zero $w^* \in \mathbb{R}^d$ and $\gamma > 0$ such that:

$$y^t \leq w^*, x^t \geq \gamma \implies \langle w^*, x^t \rangle > \gamma \text{ if } y^t = +1$$
$$\langle w^*, x^t \rangle < -\gamma \text{ if } y^t = -1$$

Convergence of perceptron happens under linear separability assumption

no weight update after finite number of mistakes made by the perceptron

$\rightarrow$ To run for a very large number of rounds — $T \rightarrow$ The perceptron makes M mistakes in these T rounds $\rightarrow$ To show: $M < \infty, M \leq B$

Consider a round t at which perceptron makes a mistake.

$$\hat{y}^t \neq y^t$$

$$w^{t+1} = w^t + y^t x^t \quad \text{before mistake}$$

$\langle w^*, w^{t+1} \rangle$ and $\langle w^*, w^t \rangle$ behave?

$$\langle w^{t+1}, w^* \rangle = \langle w + y^t x^t, w^* \rangle = \langle w, w^* \rangle + \langle y^t x^t, w^* \rangle = \langle w, w^* \rangle + y^t \langle x^t, w^* \rangle$$

$\vec{u}, \vec{v}, \vec{z}.$

$$\langle k\vec{u} + \beta\vec{v}, \vec{z} \rangle = k\langle \vec{u}, \vec{z} \rangle + \beta\langle \vec{v}, \vec{z} \rangle$$

$$\langle w^{t+1}, w^* \rangle - \langle w^t, w^* \rangle$$

$$= y^t \langle x^t, w^* \rangle > \gamma$$

$$\underbrace{\langle w^{t+1}, w^* \rangle - \langle w^t, w^* \rangle}_{>\gamma}$$

$$\sum_{t=0}^{T} \langle w^{t+1}, w^* \rangle - \langle w^t, w^* \rangle$$